Sahaj Bhadja

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Summary

Enterprise software engineer with 2+ years enhancing **production applications, SQL Server systems, and automated workflows**; experienced in performance, resiliency, high-volume transactions, defect resolution, AI integrations, testing, architecture, APIs, and documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **production services and relational data workflows** across SQL Server, Azure SQL, PostgreSQL, and FastAPI, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Automated 5,000+ recurring monthly manual business workflows with validation, reconciliation, explicit states, and failure recovery, sustaining 99%+ reliability while improving throughput 3x across high-volume processing services.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **reliable customer-focused React interfaces and Flask APIs** for uploads, dashboards, processing, review, and feedback, serving 150+ users and 10,000+ daily requests with stable production performance.
- Integrated OpenAI, Claude, and Qwen behind configurable services, collaborating with product stakeholders, QA, and engineers through requirements, testing, code reviews, production CI/CD releases, defect resolution, and documentation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Enhanced **secure scalable production microservices and API integrations** using OOP design patterns, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, and maintainability.
- Diagnosed and resolved defects while reducing technical debt 20%, maintaining 20+ unit and integration tests, automated builds, deployment workflows, architecture documentation, and operational reports through Jenkins CI/CD.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **enterprise billing, invoice, vendor, and reporting services** within a React, Node.js, and Azure SQL architecture, securely synchronizing 2M+ records across eight modular business systems and workflows.
- Improved production performance, resiliency, and scalability through caching, automated tests, Docker, Jenkins, and monitoring, achieving 95%+ build validation while reducing deployment time 50% and resource waste 42%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **financial transaction workflows** for pricing, payments, coupons, refunds, disputes, and financial status reporting, supporting 50,000+ records through Flask APIs, Firebase synchronization, validation, and sub-200-millisecond responses.
- Documented system architecture, APIs, and 12+ integration flows while automating webhook-driven transaction updates, reducing manual refund handling 70% and improving operational efficiency across mobile and web applications.

Skills

Languages: SQL, Python, Java, JavaScript, TypeScript, Bash
Applications: FastAPI, Flask, Node.js, React.js, REST APIs, microservices, system architecture
Databases: SQL Server, Azure SQL, PostgreSQL, MySQL, schema design, complex queries, indexing, tuning
Financial Systems: Billing, invoices, payments, transaction workflows, reconciliation, validation, reporting, automation
Delivery: AI integrations, testing, QA, debugging, Git, Jenkins, GitHub Actions, CI/CD, monitoring, documentation

Projects

FinNexus Live | *Financial Systems, TypeScript, MongoDB*

- Built a **high-volume financial transaction platform** supporting 50,000+ users and 1M+ transactions with secure bank integrations, real-time updates, advanced filtering, MongoDB persistence, and Kubernetes-based CI/CD automation.

Chat2SQL Code | *SQL Server, Java, TypeScript*

- Enhanced an **enterprise database application** supporting SQL Server and 10+ systems with schema exploration, structure synchronization, performance-aware query generation, data management, validation, Docker, Maven, and cross-platform troubleshooting workflows.

HMIS Live | *Workflow Automation, APIs, Monitoring*

- Developed a **workflow automation application** with role-based access, dashboards, appointments, REST integrations, notifications, and Sentry monitoring, halving record retrieval time while supporting 100+ daily workflows and reducing administrative work 30%.